

Recommendations for Gulf Ecosystem Restoration

Kacky Andrews

Oceans and Coasts Program Director

North America Region

June 27, 2011



Protecting nature. Preserving life.®



TEXAS A&M
UNIVERSITY
CORPUS
CHRISTI

HARTE
RESEARCH INSTITUTE
FOR GULF OF MEXICO STUDIES

The Nature
Conservancy



Protecting nature. Preserving life.™



Audubon



Ocean
Conservancy



NATIONAL
WILDLIFE
FEDERATION®

Community Resilience



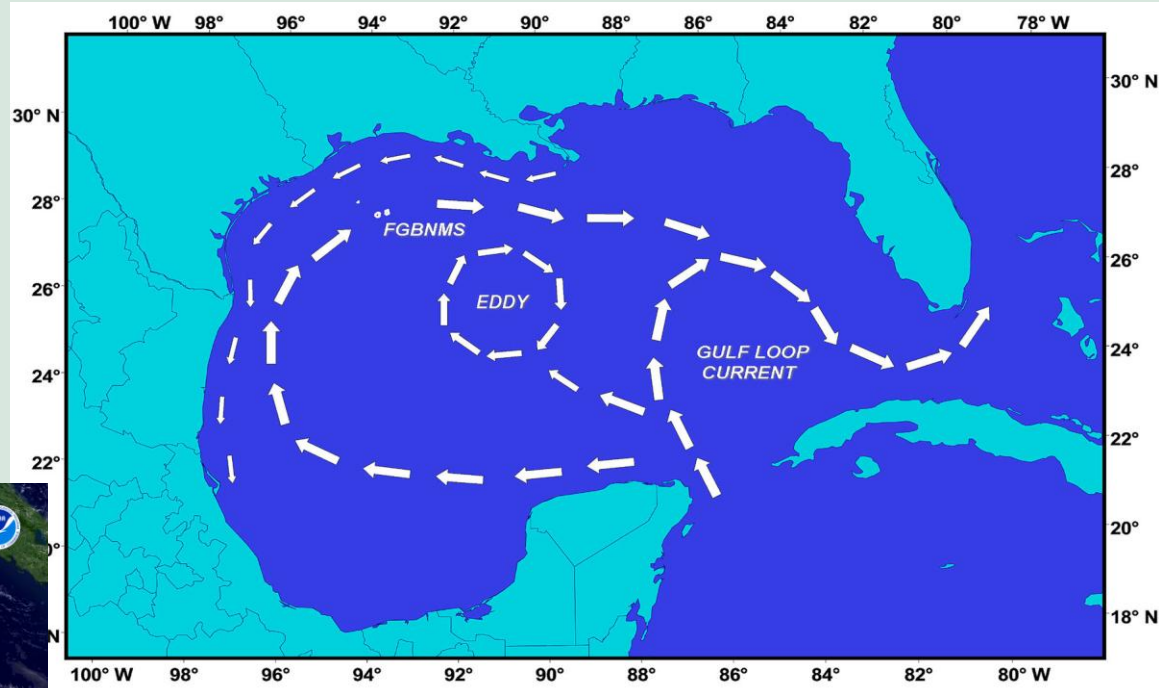
Vision for a Healthy Gulf

- Sustained biodiversity
- Productive habitats
- Healthy populations of commercial species
- Access for the public
- Fishable, swimmable beaches and rivers
- Thriving cities and towns
- Citizen stewards



Ecosystem Drivers

- Loop Current
- Freshwater Inflow
- Hurricanes and Tropical Storms
- Ecological Buffers and Filters



Impacts to the Gulf

- Agricultural runoff
- Development impacts
- Channelization and dredging of rivers
- Damaging fishing techniques
- Endocrine disruptors
- Climate change
- Harmful Algal Blooms
- Invasive species
- Nutrients
- Sea level rise



already lost in the Gulf...

50%

inland and
coastal wetlands

over

50%

oyster reefs

up to

60%

seagrass beds

up to

33%

mangrove forests

Essential Priorities

The background of the slide is a photograph of a natural landscape. In the foreground, there are tall, green reeds or grasses growing in a body of water. The water is calm, reflecting the sky and the clouds. In the middle ground, there is a dense line of trees, possibly a forest or a park. The sky is blue with scattered white and grey clouds. The overall scene is peaceful and natural.

1. Recover Habitat
2. Sustain Fish and Wildlife
3. Get the Water Right
4. Conserve Special Places

Recover Habitat - Categories

- Wetlands
- Oyster Reefs
- Seagrasses
- Barrier Islands and Beaches
- Coral Reefs
- Mangroves
- Deep Water Ocean Habitats

Recover Habitat Sample Outcomes

- Wetlands: Restore hydrological of Miss. River by capturing 80% of the sediment coming down the river
- Oyster reefs: Restore at 50% of historical area of oyster reefs Gulf-wide
- Seagrasses: Restore 500,000 acres
- Beaches: Increase percentage of beaches using best management practices

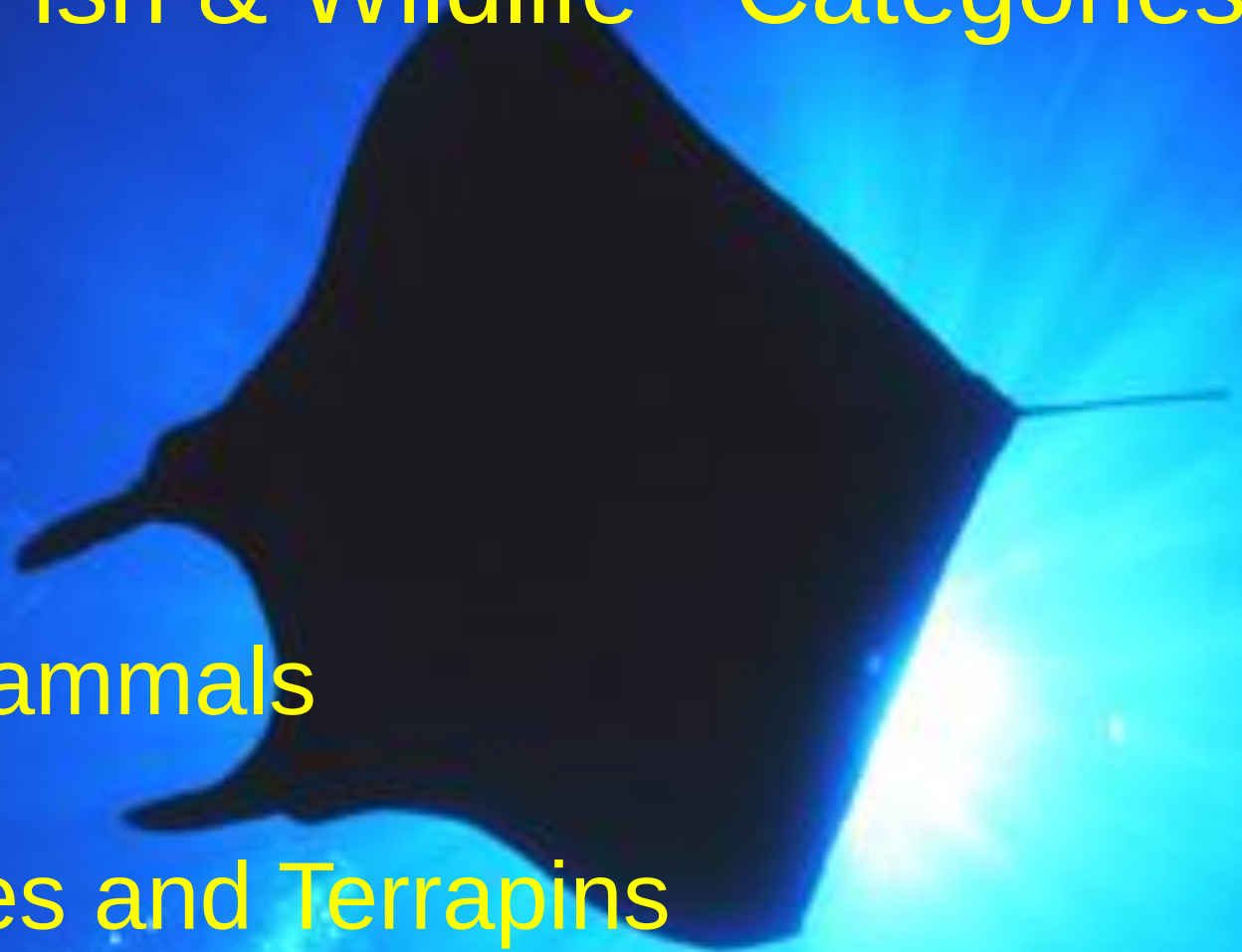
Sustain Fish & Wildlife - Categories

Fish

Birds

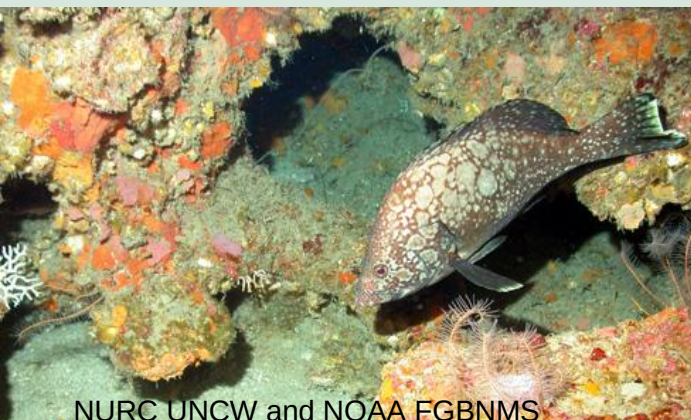
Marine Mammals

Sea Turtles and Terrapins



Fish & Wildlife Sample Outcomes

- Fish: All federally managed species are managed at optimum yield
- Birds: Protect colonially and beach nesting bird sites
- Sea turtles: Recover endangered and threatened (E & T) sea turtles sufficient to remove them from federal E & T list



Get the Water Right - Categories

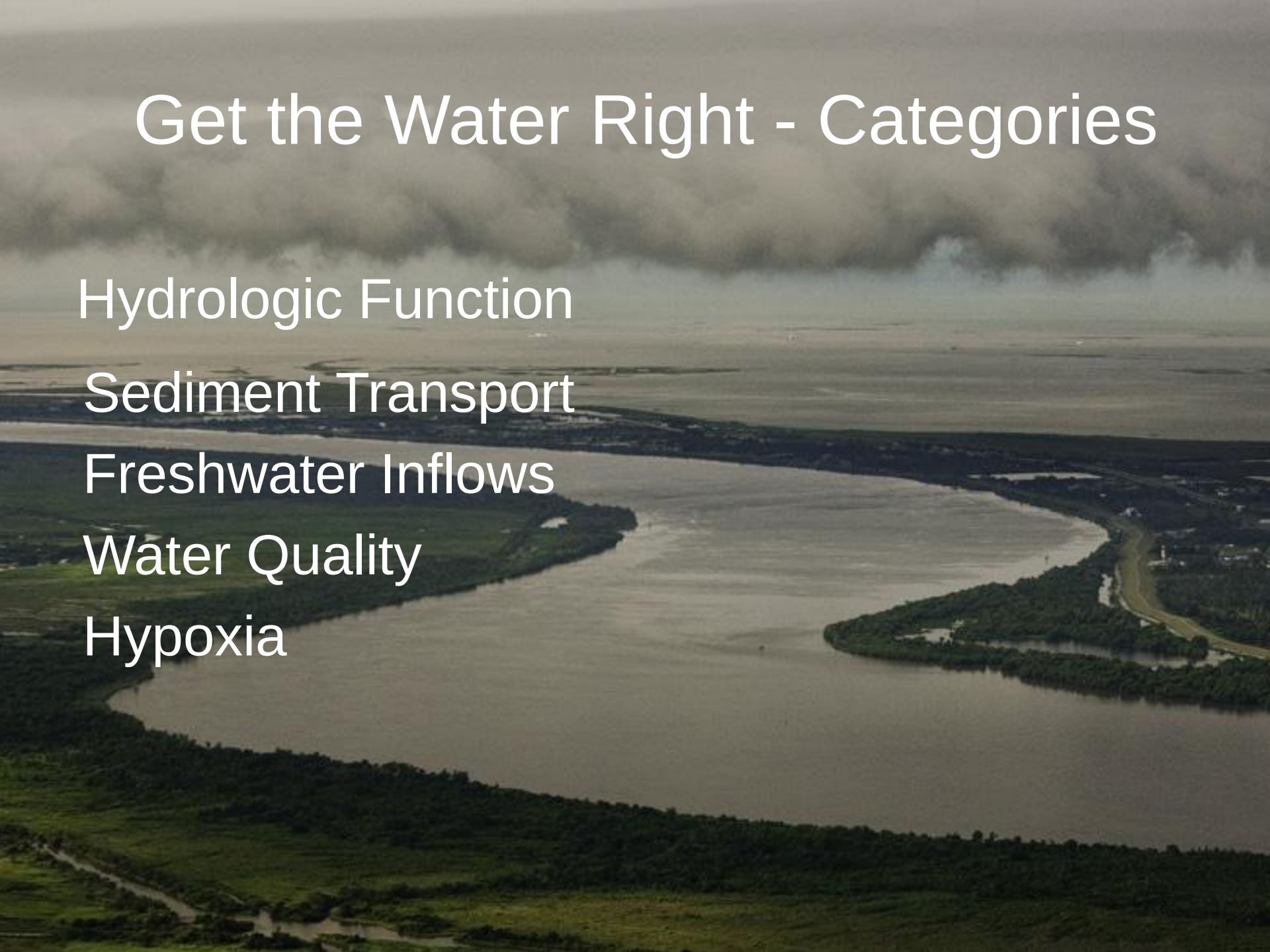
Hydrologic Function

Sediment Transport

Freshwater Inflows

Water Quality

Hypoxia



Get the Water Right

Sample Outcomes

- **Quality:** Meet water quality standards for pathogens and nutrient load reduction using TMDLs
- **Quantity:** Establish scientifically sound flow regimes for each bay and estuary
- **Hydrologic Function:** Improve sediment transport to reduce rate of wetland loss

Conserve Special Places

- Ensure connectivity of special places to conserve biodiversity
- Sample Outcomes:
 - Form and empower local and regional partnerships to promote conservation of special places
 - Identify key areas of ecological connectivity and promote their conservation

Supporting Activities

- Develop Environmental Report Card and Long-term Monitoring Plan
- Strengthen Science-based Adaptive Management
- Incorporate Impacts from Climate Change
- Develop Market-based Solutions to Help Protect Value of Nature
- Promote Citizen Science and Stewardship through Environmental Education



• ADVANCING SCIENCE • BUILDING PARTNERSHIPS • CELEBRATING DIVERSITY • EXPANDING REACH • SETTING PRIORITIES • PROTECTING NATURE • PRESERVING LIFE • TAKING ACTION •



a national
destination



productive fisheries





an economic
engine





Gulf of Mexico Restoration Decision Support



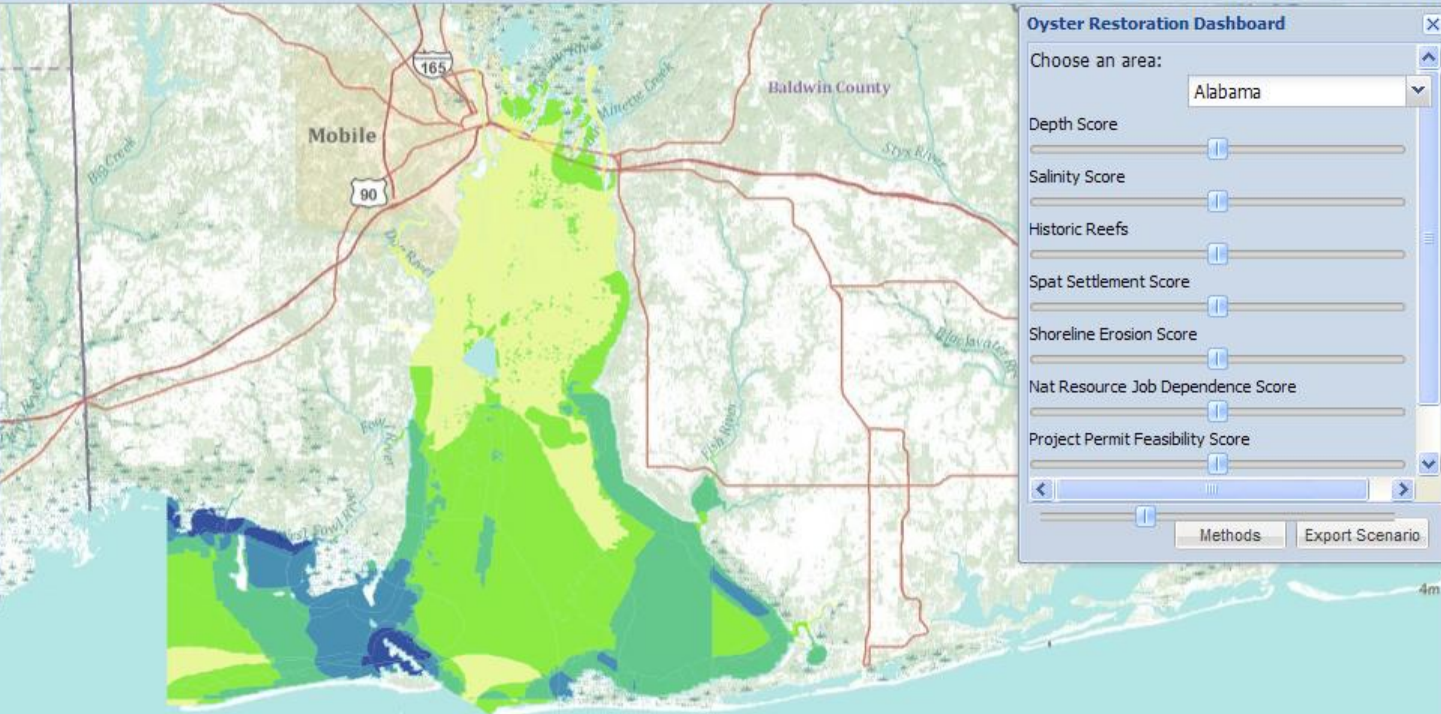
Map Layers | Legend | Change to Split View | Guide | Restoration Dashboard

Map Layers

Left Map | Right Map

Clear All | Synch Maps | Set Transparency

- Florida
- Alabama
 - Oyster Restoration Site Selection
 - Restoration Projects
 - Oyster Reefs
 - Bathymetry
 - Salinity
 - Submerged Vegetation
 - Coastal Access
 - Coastal Hazards
 - Social and Economic
- Mississippi
- Louisiana
- Texas
- Gulf of Mexico
 - Ecological
 - Socioeconomic



Oyster Restoration Dashboard

Choose an area:
Alabama

Depth Score

Salinity Score

Historic Reefs

Spat Settlement Score

Shoreline Erosion Score

Nat Resource Job Dependence Score

Project Permit Feasibility Score

Methods | Export Scenario

LAT: 30.494 N LON: 88.195 W

[The Nature Conservancy](#) | [Coastal Resilience](#) | [Legal Disclosure](#)

Gulf of Mexico Restoration Decision Support



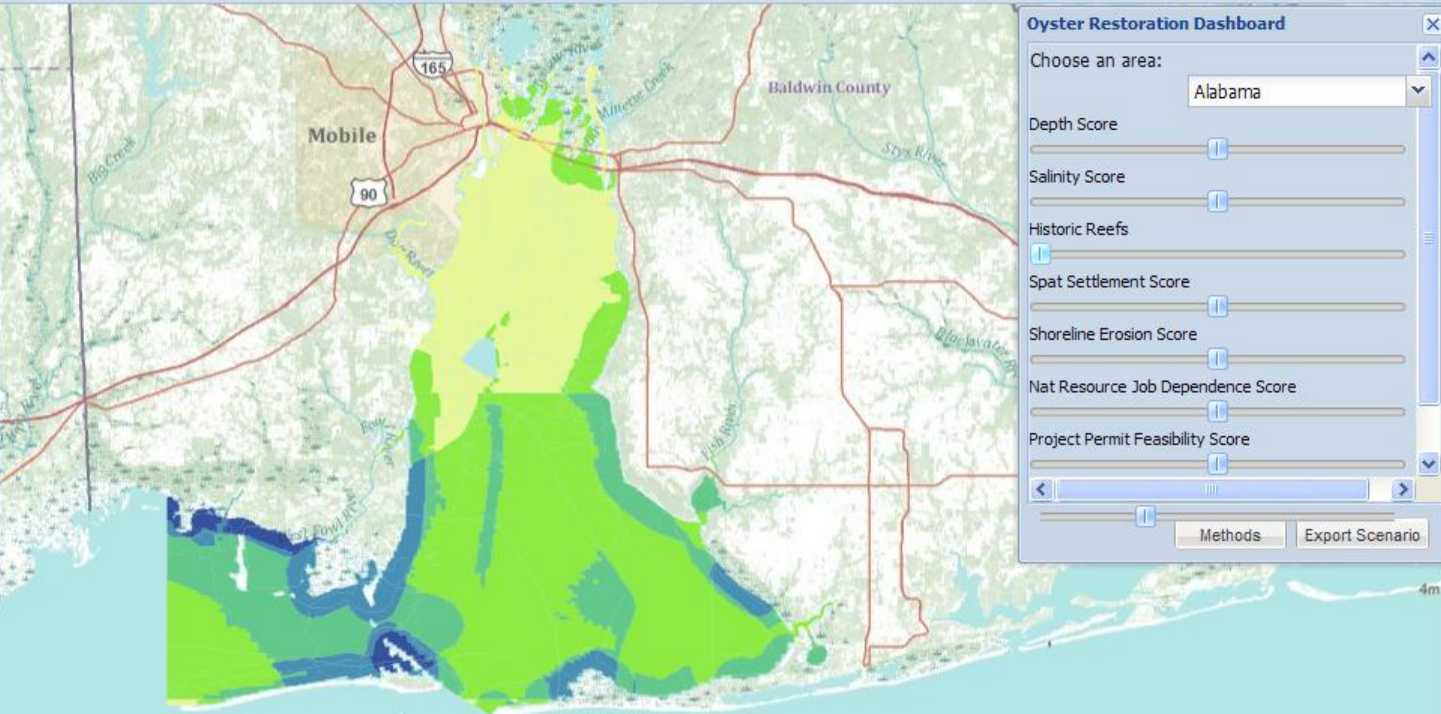
Map Layers | Legend | Change to Split View | Guide | Restoration Dashboard

Map Layers

Left Map | Right Map

Clear All | Synch Maps | Set Transparency

- Florida
- Alabama
 - Oyster Restoration Site Selection
 - Restoration Projects
 - Oyster Reefs
 - Bathymetry
 - Salinity
 - Submerged Vegetation
 - Coastal Access
 - Coastal Hazards
 - Social and Economic
- Mississippi
- Louisiana
- Texas
- Gulf of Mexico
 - Ecological
 - Socioeconomic



Oyster Restoration Dashboard

Choose an area:
Alabama

Depth Score

Salinity Score

Historic Reefs

Spat Settlement Score

Shoreline Erosion Score

Nat Resource Job Dependence Score

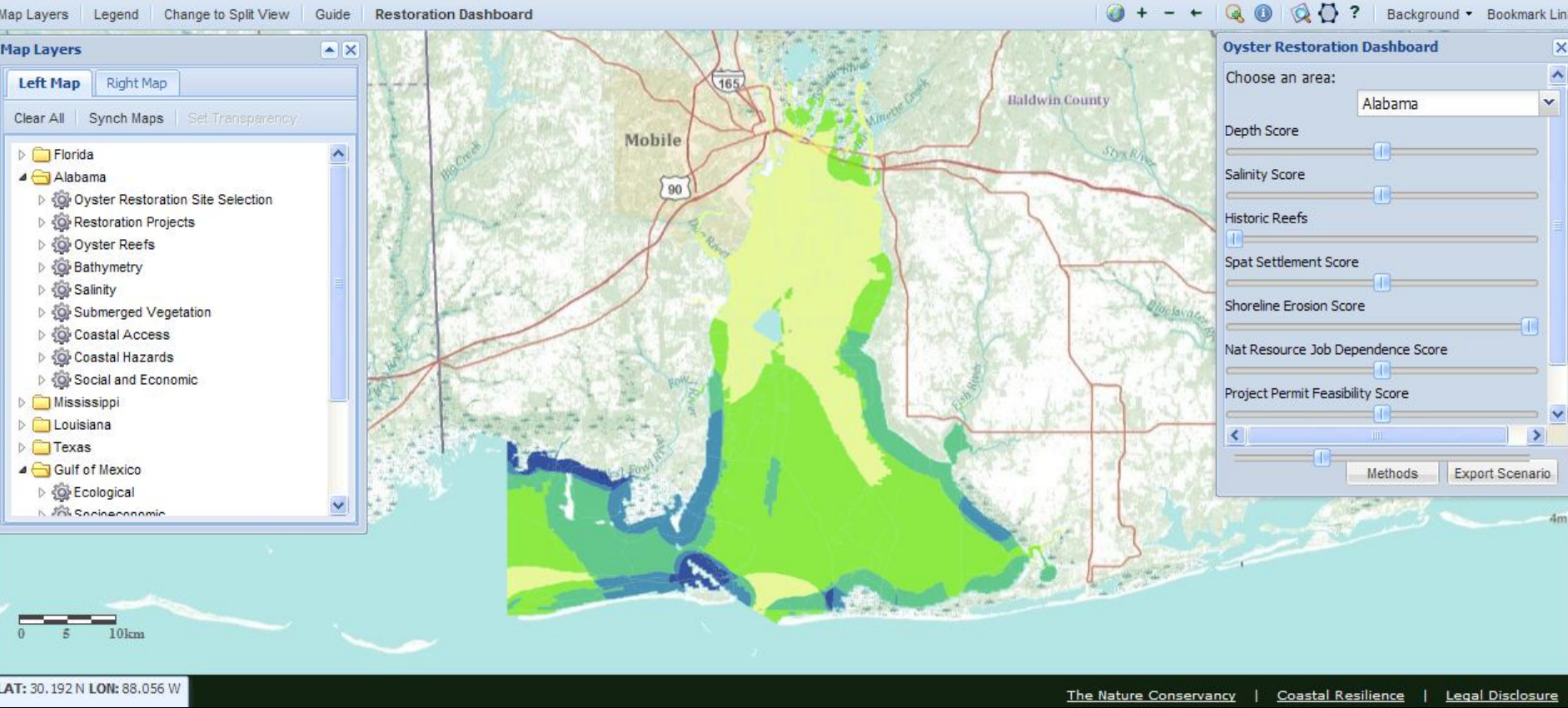
Project Permit Feasibility Score

Methods | Export Scenario

0 5 10km
LAT: 30.285 N LON: 87.577 W

The Nature Conservancy | Coastal Resilience | Legal Disclosure

Gulf of Mexico Restoration Decision Support



LAT: 30.192 N LON: 88.056 W

Gulf of Mexico Restoration Decision Support



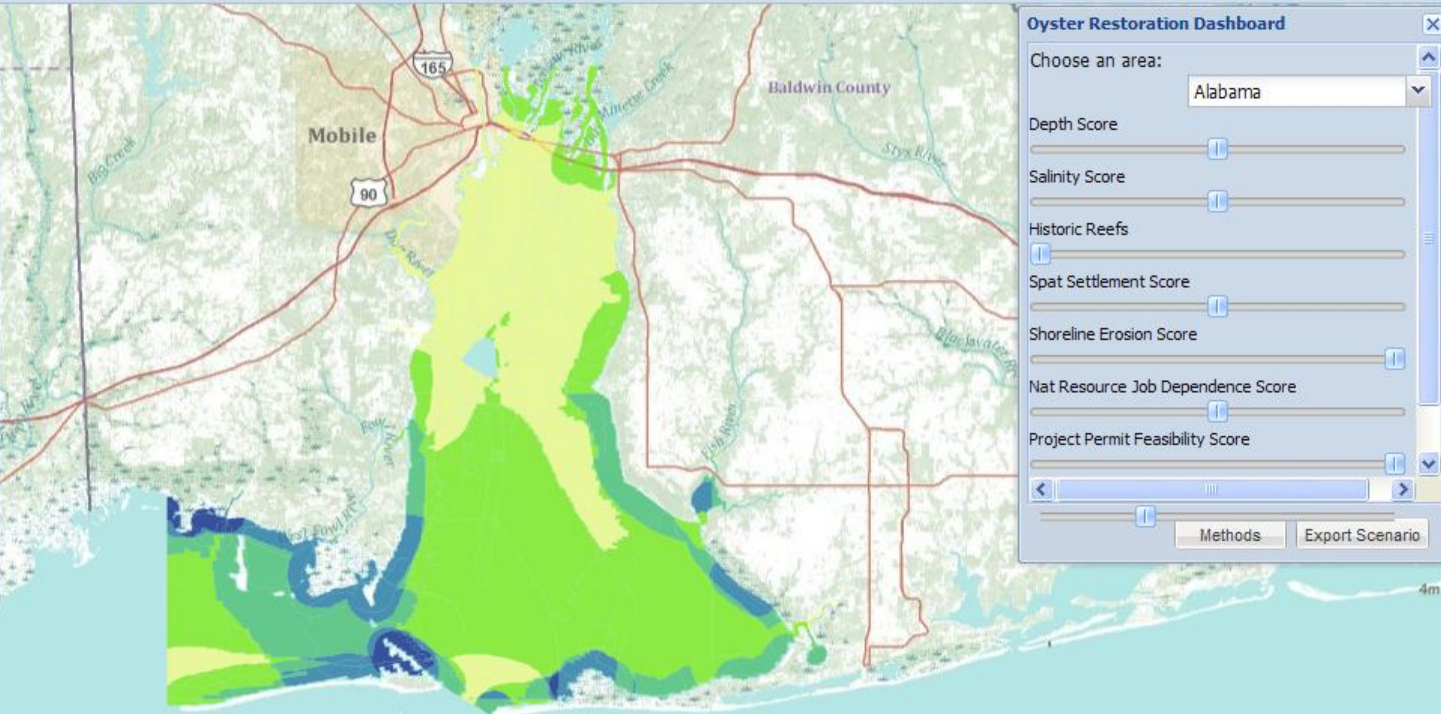
Map Layers Legend Change to Split View Guide Restoration Dashboard

Map Layers

Left Map Right Map

Clear All Synch Maps Set Transparency

- Florida
- Alabama
 - Oyster Restoration Site Selection
 - Restoration Projects
 - Oyster Reefs
 - Bathymetry
 - Salinity
 - Submerged Vegetation
 - Coastal Access
 - Coastal Hazards
 - Social and Economic
- Mississippi
- Louisiana
- Texas
- Gulf of Mexico
 - Ecological
 - Socioeconomic



Oyster Restoration Dashboard

Choose an area:
Alabama

Depth Score

Salinity Score

Historic Reefs

Spat Settlement Score

Shoreline Erosion Score

Nat Resource Job Dependence Score

Project Permit Feasibility Score

Methods Export Scenario

LAT: 30.642 N LON: 87.674 W

The Nature Conservancy | Coastal Resilience | Legal Disclosure

Gulf of Mexico Restoration Decision Support



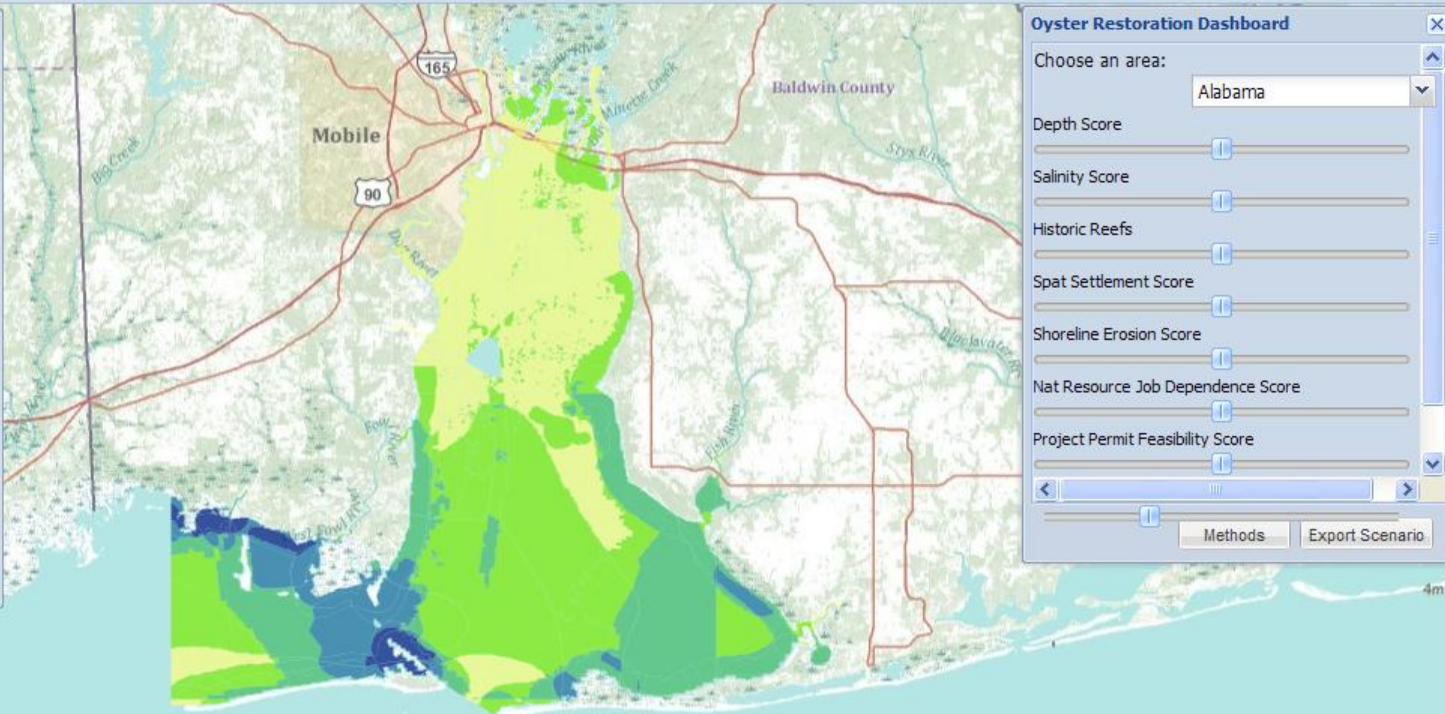
Map Layers | Legend | Change to Split View | Guide | Restoration Dashboard

Map Layers

Left Map | Right Map

Clear All | Synch Maps | Set Transparency

- Florida
- Alabama
 - Oyster Restoration Site Selection
 - Restoration Projects
 - Oyster Reefs
 - Bathymetry
 - Salinity
 - Submerged Vegetation
 - Coastal Access
 - Coastal Hazards
 - Social and Economic
- Mississippi
- Louisiana
- Texas
- Gulf of Mexico
 - Ecological
 - Socioeconomic



Oyster Restoration Dashboard

Choose an area:
Alabama

Depth Score

Salinity Score

Historic Reefs

Spat Settlement Score

Shoreline Erosion Score

Nat Resource Job Dependence Score

Project Permit Feasibility Score

Methods | Export Scenario

LAT: 30.494 N LON: 88.195 W

The Nature Conservancy | Coastal Resilience | Legal Disclosure